

## Italian-Maturita-2026-04-14-English

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### Simulation of the Mathematics Portion of the Maturità Exam

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La Scuola d'Italia Guglielmo Marconi – New York

Date: 14 April 2026

Surname and Name: \_\_\_\_\_

#### For the Scientific High School Track

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Answer 4 of the following questions:

1. Consider the following proposition:

*“If two solids have equal volume, then, when cut by a family of parallel planes, they determine cross-sections of equal area on each of them.”*

State whether the statement is true or false, and give a fully justified explanation for your answer.

2. Recalling that the side of a regular decagon inscribed in a circle is equal to the chord subtending an arc in a circle of radius ( $r$ ), prove that

[ ]

3. Among cylindrical containers having the same surface area ( $S$ ) (the lateral area plus the base), which one has the maximum volume?

4. State the rule of the Marquis de L'Hôpital (1661–1704) and apply it to prove that [ ]

5. Determine a third-degree polynomial ( $P(x)$ ) such that: [ ]

6. If [ ] with ( $n > 3$ ) are in arithmetic progression, what is the value of ( $n$ )?

7. Determine, as ( $k$ ) varies, the number of real solutions of the equation: [ ]

8. Let the function be defined by [ ] Specify the domain of ( $f$ ), and determine the sign of the first and second derivatives at the point ( $x=$ ).

9. Let [ ] Does the limit [ ] exist? Justify your answer.

10. According to the highway code, the “steep ascent” sign (see figure) warns drivers of a stretch of road with a slope large enough to be dangerous.

The slope is expressed as a percentage, and in the example it is 10%.

If a straight road is built that, over a distance of 1.2 km, rises by 85 m in elevation, what is its inclination (in degrees)? What percentage should be shown on the sign?

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**Maximum duration of the test:** 2 hours.

The use of scientific and/or graphing calculators is allowed, provided they do not have symbolic computation capability.